Resonance-based Tossing Control for Bulk Materials Transportation of an Overhead Crane

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***Abstract*—Transportation of bulk materials using an overhead crane system possesses a special property, that is, the transferred materials can be dropped/discharged while in the air. In order to exploit such a unique feature, this paper introduces a new concept, named tossing control methodology, to enhance transportation productivity. A specific type of tossing controller is proposed, which relies on the phenomenon of linear resonance to induce oscillation in periodically increasing amplitude. A novel resonance-based tossing controller—comprising an optimal linearization law and a resonance evoking component—is formulated. It will be shown that, under similar requirements of bulk materials transportation, conditions, and actuator constraints, the resonance-based tossing control can reduce the transferring time up to 26.5% compared with the well-known minimum-time vibration controller—the fastest member of the vibration control group. The results show that, although it is not always able to obtain a faster transportation using the resonance tossing controller, evidence for breaking the time limitation of the minimum-time vibration control is provided. Both simulation and experimental results are presented to verify the effectiveness of the proposed resonance tossing control method, and possible future developments are discussed.**

***Index Terms*—Bulk materials unloading, crane control, tossing control, resonance control, vibration excitation, motion planning, underactuated system.**

I. Introduction

OVERHEAD cranes are widely used in various places such as in factories, storage yards, harbors, etc. to transfer heavy payloads point-to-point. The payload is suspended under the trolley by wire ropes and its height can be adjusted by the hoisting motion. The trolley is actuated to relocate the payload within the workspace. The payloads are usually large blocks or bulk materials (such as sand, cement, etc.). When transferring the bulk materials, the overhead crane must utilize a specially designed grab bucket which can close/open to effectively grasp/drop the materials (see Fig. 1). The fundamental property of the bulk materials transportation is that the materials can be dropped/discharged from the air and they naturally fall down toward their intended destination by gravity. This is completely different to a large block transfer, in which the payload must be eventually laid down on the ground by the overhead crane. It is well-known that the payload exhibits vibration during and after transportation [1]. The residual oscillation is usually undesirable when transferring a large block payload, since, for safety, the swinging motion needs to cease before the payload can be safely landed. Therefore, in such circumstances, vibration (suppression) control (which is plentiful in the literature [1]–[15]) must be emphasized. In the context of bulk materials transportation, its aforementioned unique property, namely that materials can be discharged while in the air, implies a favorable consequence, that is, the zero terminal swing angle constraint as found with the vibration control can be relaxed. As a result, the requirements of bulk materials transportation are stated as follows: the payload (grab bucket with materials inside) should be transferred above the discharging destination, at which the payload's horizontal and vertical velocities are zero, meanwhile the trolley must stop[1], and the terminal swing angle is not necessarily zero. The relaxation of the constraint on the zero terminal swing angle indeed offers the possibility of further reducing the

Fig. 1. An overhead crane is transferring bulk material in a warehouse.

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[1]These requirements are explained as follows. At the discharging destination, the payload speed in both horizontal and vertical directions must be zero in order to make use of the following advantages. Firstly, it ensures that, in ideal conditions, when the grab opens, the bulk materials fall down along the vertical axis, which prevents spilling. Secondly, the control input for the returning phase can be easily obtained by simply applying the designed control input of the transferring phase backward in time (see Section III-C). Lastly, after discharging, the payload tendency is to swing toward the initial position, which naturally makes the returning phase more time-efficient. In addition, the trolley velocity must also be zero since after discharging the material, to save time, the trolley needs to immediately return to the initial position to supply material for the next transportation cycle.

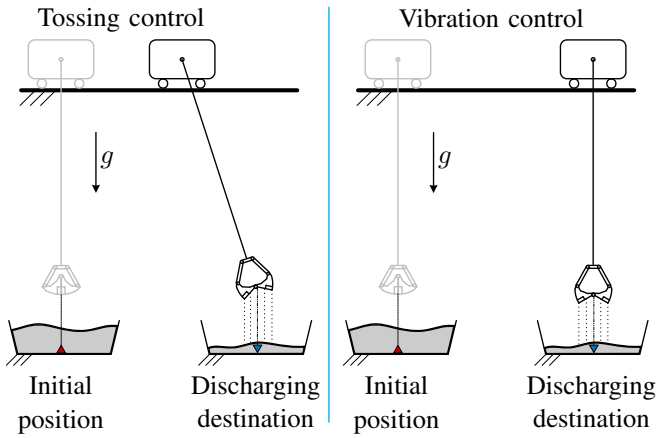

Fig. 2. Difference between tossing control and vibration control.

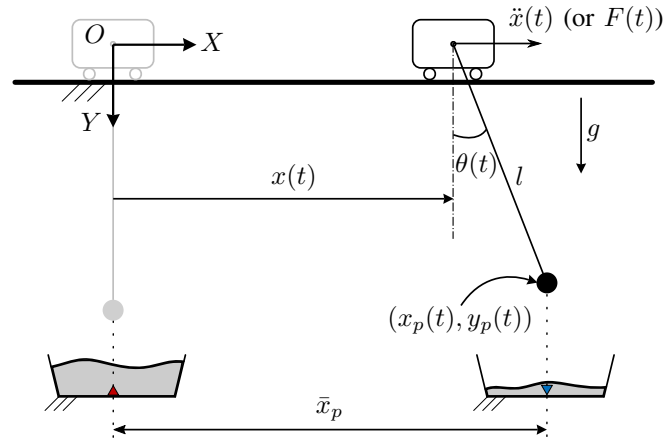

Fig. 3. Mathematical model of the overhead crane system.

transferring time by exciting the payload vibration rather than suppressing/minimizing it. Such a type of control is termed tossing control. Note that, both tossing and vibration control methodologies can realize the requirements of bulk materials transportation. The only difference between them is the tossing control aims at achieving a large terminal swing angle, whereas the swing angle must be zero in the vibration control. This dissimilarity is illustrated in Fig. 2.

This paper will introduce a resonance-based tossing control for an overhead crane system. The proposed method utilizes the linear resonance phenomenon (see [16, p. 113], [17, p. 162]) to generate a large swing angle with periodically increasing amplitude. In the paper, the linear resonance is employed instead of the nonlinear one due to the following reasons. First, the linear vibration frequency can be exactly computed, whereas it is generally not possible in the case of the nonlinear vibration [18]. Second, the linear resonance frequency is independent from the magnitude of control input and oscillation itself, whereas such a property may not hold in the nonlinear resonance [19]. Therefore, by using the linear resonance, the controller design can be facilitated. The resonance control law is given by a parameterized composite sine and cosine function, whose frequency is identical with that of the system. By that means, the resonance can be created. Since a large swing angle is supposed to be generated, the nonlinear model of the overhead crane ought to be used. Therefore, in order to obtain the linear resonance for a nonlinear system, an exact linearization law is added to the resonance control input. The technical difficulty in using such a technique is that the control effort for linearizing the system grows rapidly as the swing angle increases, which is even unbounded when the swing angle approaches the singular points at $\pm\pi/2$ if a conventional linearization law is used. To solve that problem, this paper will introduce an optimal gain for the linearization law to minimize its magnitude. A favorable property of the proposed optimal exact linearization law is that its magnitude can be bounded by a known limit for all swing angles in $(-\pi/2,\ \pi/2)$. Essential terminal conditions resulted from the tossing control requirements are employed to determine the unknown parameters of the total resonance control law. Both maximum velocity and maximum acceleration of the actuator are taken into account. To the best of authors' knowledge, none of existing researches investigated a similar problem relating to the tossing control methodology.

It is well-known that, among all vibration controllers, the minimum-time (or time-optimal) control universally provides the smallest possible transferring time [4], [20]–[22]. Nevertheless, at its best, the proposed resonance-based tossing controller is able to save 26.5% transferring time (or 6.4 seconds in one cycle of transportation, see Section IV) compared with the minimum-time vibration control. Therefore, this evidence confirms the main aim of the paper and proves the possibility of breaking the time limitation of the minimum-time vibration control in the context of bulk materials transportation.

The remainder of the paper is organized as follows. In Section II, a mathematical model of the overhead crane system is presented. Moreover, requirements of the bulk materials transferring process are introduced. Section III is devoted to the design of the resonance-based tossing controller to fulfill the transportation task. Comparative simulation result of the proposed tossing controller and the minimum-time vibration control is provided in Section IV. Experimental result on a small-scale overhead crane is given in Section V to verify the effectiveness of the proposed motion planning method. Possible future work is elaborated in Section VI. Finally, Section VII draws conclusions of the paper.

II. Problem formulation

Swinging motion of the payload resulted from the trolley actuation is illustrated in Fig. 3, in which $x(t)$ (m) and $\theta(t)$ (rad) are the trolley position and the swing angle. The net force driving the trolley is denoted as $F(t)$. Furthermore, m (kg), M (kg), l (m), and g (= 9.81 m/s^2) represents the payload mass, trolley mass, rope length, and gravitational acceleration respectively. The following assumptions are made to establish the dynamical model of the overhead crane system: the payload is considered as a point mass, the rope is assumed to be a rigid rod, and the air resistance to the payload could be negligible. Note that, the rigid rope assumption is reasonable and it is commonly used in the literature due to the fact that the elasticity modulus of the wire rope is very high. In other words, the axial elasticity effect of the rope is minor and could be negligible. The Euler–Lagrange method will be employed

to establish the system dynamic. From Fig. 3, horizontal and vertical positions of the payload can be computed by

$$x_p(t) = x(t) - l\sin\theta(t) \tag{1}$$

$$y_p(t) = l\cos\theta(t). \tag{2}$$

By differentiating (1) and (2), horizontal and vertical velocities of the payload are

$$\dot{x}_p(t) = \dot{x}(t) - l\dot{\theta}(t)\cos\theta(t) \tag{3}$$

$$\dot{y}_p(t) = -l\dot{\theta}(t)\sin\theta(t). \tag{4}$$

The total kinetic energy of the system comprises translation energies of the trolley and the payload, thus it can be written by

$$\mathcal{K} = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left(\dot{x}_p^2 + \dot{y}_p^2\right). \tag{5}$$

The potential energy of the system is given as $\mathcal{V} = -mgy_p$. Therefore, the Lagrangian function can be established by

$$\mathcal{L} = \mathcal{K} - \mathcal{V} = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left(\dot{x}_p^2 + \dot{y}_p^2\right) + mgl\cos\theta. \tag{6}$$

The Euler–Lagrange equation applying to the actuated coordinate $x(t)$ and the unactuated coordinate $\theta(t)$ reads:

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{x}}\right) - \frac{\partial\mathcal{L}}{\partial x} = F. \tag{7}$$

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) - \frac{\partial\mathcal{L}}{\partial\theta} = 0. \tag{8}$$

By developing (7) and (8) using (3), (4) and (6), the mathematical model of the overhead crane system is governed by

$$(M+m)\ddot{x} - ml\ddot{\theta}\cos\theta + ml\dot{\theta}^2\sin\theta = F \tag{9}$$

$$l\ddot{\theta} + g\sin\theta = \cos\theta\ddot{x} \tag{10}$$

From (9) and (10), it is recognized that one can either use $F(t)$ or $\ddot{x}(t)$ as the control input. For the motion planning task, $\ddot{x}(t)$ is the more appropriate choice for the following reasons: i) the problematic frictional forces presented in $F(t)$ can be avoided, ii) the first and second integration of $\ddot{x}(t)$ will result in the velocity $\dot{x}(t)$ and the position $x(t)$ commands, which can be easily realized by ubiquitous servo drives, iii) the system dynamic is only dependent on the rope length l, which can be easily measured in practice. Therefore, the dynamical equation (10) is employed in the control design, in which the control input is $\ddot{x}(t)$. Note that, by choosing such a control signal, it is assumed that the servo motor can satisfactorily track the planned reference trajectory, which is an ordinary requirement for most industrial servo drives. Moreover, due to physical limitation of the trolley actuator, the following maximum velocity and maximum acceleration constraints must be strictly complied:

$$|\dot{x}(t)| \le v_{\max} \text{ and } |\ddot{x}(t)| \le a_{\max}, \tag{11}$$

where $v_{\max} > 0$ and $a_{\max} > 0$. By using the information of the servo motor's maximum velocity and maximum torque, as well as the structure of the transmission system, the values of $v_{\max}$ and $a_{\max}$ can be easily estimated [23].

Throughout the paper, it is assumed that the payload starts from rest, namely $x(0) = \dot{x}(0) = \theta(0) = \dot{\theta}(0) = 0$ [2], [4], [20], [22], [24]–[26]. In order to realize the tossing control requirements (see Section I), the following terminal conditions must be fulfilled

$$\dot{x}(t_f) = \dot{x}_p(t_f) = \dot{y}_p(t_f) = 0,\ x_p(t_f) = \bar{x}_p,\ \theta(t_f) < 0. \tag{12}$$

In (12), t_f is the terminal time and $\bar{x}_p$ represents the horizontal desired destination of the payload (see Fig. 3). By imposing (12), the payload is transferred to the discharging destination ($x_p(t_f) = \bar{x}_p$) at zero velocities in both horizontal and vertical directions ($\dot{x}_p(t_f) = \dot{y}_p(t_f) = 0$), the trolley stops ($\dot{x}(t_f) = 0$), and a large terminal swing angle is obtained ($\theta(t_f) < 0$). From (3) and (4), horizontal and vertical velocities of the payload at the terminal time t_f are

$$\dot{x}_p(t_f) = \dot{x}(t_f) - l\dot{\theta}(t_f)\cos\theta(t_f) \tag{13}$$

$$\dot{y}_p(t_f) = -l\dot{\theta}(t_f)\sin\theta(t_f). \tag{14}$$

In order to realize the condition $\dot{x}(t_f) = \dot{x}_p(t_f) = \dot{y}_p(t_f) = 0$ of (12), we should make $\dot{x}(t_f) = \dot{\theta}(t_f) = 0$. If this condition is strengthened by adding $\theta(t_f) = 0$, it will become the vibration control, which is not of interest in this paper. Moreover, by using (1), the remaining condition of (12), namely $x_p(t_f) = \bar{x}_p$, can be developed to $x(t_f) - l\sin\theta(t_f) = \bar{x}_p$. In summary, the terminal conditions (12) are converted to

$$\dot{x}(t_f) = \dot{\theta}(t_f) = 0,\ x(t_f) - l\sin\theta(t_f) = \bar{x}_p,\ \theta(t_f) < 0. \tag{15}$$

The following section will be devoted to formulating the resonance tossing control to realize (15) while ensuring (11).

III. Design of Resonance-Based Tossing Control

A. Resonance control with an optimal linearization law

The basic idea of the proposed method (to realize the tossing control requirements) is based on the resonance phenomenon. Mechanical resonance can be evoked by matching the frequency of the control input to that of the oscillation. The main technical difficulty of utilizing the resonance for a nonlinear system is that the frequency of the nonlinear vibration strongly depends on the control input as well as the oscillation amplitude [18], [19]. This creates a dilemma in the control design process, since to generate the resonance, the control input must know the oscillation frequency in advance so that the frequency matching can be done. Furthermore, the calculation of exact vibration frequency for a nonlinear system is not an easy task, or sometimes, is impossible [18]. Therefore, in this paper, the linear resonance is employed instead of the nonlinear one. In order to generate the linear resonance phenomenon for the nonlinear system (10), the following exact linearization resonance control law is proposed

$$\ddot{x}(t) = \underbrace{g\left(\tan\theta - \frac{k\theta}{\cos\theta}\right)}_{:=L(t)} \overbrace{-\frac{a}{\cos\theta}\sin(\omega_0 t) - \frac{b}{\cos\theta}\cos(\omega_0 t)}^{:=R(t)}. \tag{16}$$

In (16), $L(t)$ is the linearization law to exactly linearize the nonlinear system (10) and $R(t)$ is the control input for

generating the resonance. The unknown parameters a, b, and ω_0 will be determined in the subsequent deductions by solving a system of nonlinear equations (see (39)). Furthermore, the constant k must be designed to minimize the magnitude of the linearization law $L(t)$. This optimal design objective is explained as follows. The methodology of the tossing control is to excite a large swing angle, which entirely depends on the magnitude of $R(t)$. As the swing angle $\theta(t)$ increases, nevertheless, the control effort for linearizing the system (or the magnitude of $L(t)$) rapidly increases (see Fig. 4 with $k = 1$). Moreover, the control input is limited by $|\ddot{x}(t)| \leq a_{\max}$. Thus, no room is left for the (prioritized) resonance generation component $R(t)$. For this reason, maximum value of $|L(t)|$ over a range of operating swing angle $\left(\theta \in \left[-\bar{\theta}, \bar{\theta}\right], \bar{\theta} > 0\right)$ needs to be minimized by k. Therefore, k should be a minimizer of the following minimax optimization problem

$$\begin{aligned} \min_{k} \quad & \max_{\theta} \left[f(k,\theta) = \left(\tan\theta - \frac{k\theta}{\cos\theta} \right)^2 \right] \\ \text{s.t.} \quad & \theta \in \left[-\bar{\theta}, \bar{\theta}\right] \text{ with } \bar{\theta} \in [0, \pi/2) \\ & k \in (0, +\infty). \end{aligned} \tag{17}$$

Note that, $\bar{\theta}$ is treated as an unknown, which can be solved together with the parameters a, b, and the terminal time t_f (see (39)). The optimal solution of (17) is given by Theorem 1.

Theorem 1. The minimizer $k^\star$ of the minimax optimization problem (17) is given by

$$k^\star = \frac{1}{\cos\theta^\star + \theta^\star \sin\theta^\star}, \tag{18}$$

where $\theta^\star$ is determined by

$$\sin\left(\bar{\theta} + \theta^\star\right) - \theta^\star \cos\left(\bar{\theta} + \theta^\star\right) = \bar{\theta}. \tag{19}$$

Proof of Theorem 1 is given in Appendix A. The optimal solution of (17) can be found by solving the nonlinear equation (19). By using the Intermediate Value Theorem [27, p. 52], it is easy to show that, for all $\bar{\theta} \in [0,\ \pi/2)$, (19) has one and only one root $\theta^\star \in \left[0,\ \bar{\theta}\right]$. Therefore, (19) can be solved by any one-dimensional root-finding solver. For instance, if a bisection method is employed, the computational complexity for solving the nonlinear equation (19) is $\mathcal{O}(n)$ [28] and so is the minimax program (17).

Denote $L^\star(t) := L(t)|_{k=k^\star}$ as the optimal linearization law. Some important properties of the optimal solution are presented in the following corollary of Theorem 1.

Corollary 1. The optimal solution (18)–(19) of the minimax program (17) has the following properties:

1) $\lim\limits_{\bar{\theta}\to(\pi/2)^-} \theta^\star = \pi/2.$
2) $\lim\limits_{\bar{\theta}\to(\pi/2)^-} k^\star = 2/\pi.$
3) $\lim\limits_{\bar{\theta}\to(\pi/2)^-} \max\limits_{\theta\in[-\bar{\theta},\bar{\theta}]} |L^\star(t)| = 2g/\pi.$

Proof. Taking the limit as $\bar{\theta} \to (\pi/2)^-$ for both sides of (19) results in

$$\lim_{\bar{\theta}\to(\pi/2)^-} \left[\sin\left(\bar{\theta} + \theta^\star\right) - \theta^\star \cos\left(\bar{\theta} + \theta^\star\right)\right] = \pi/2. \tag{20}$$

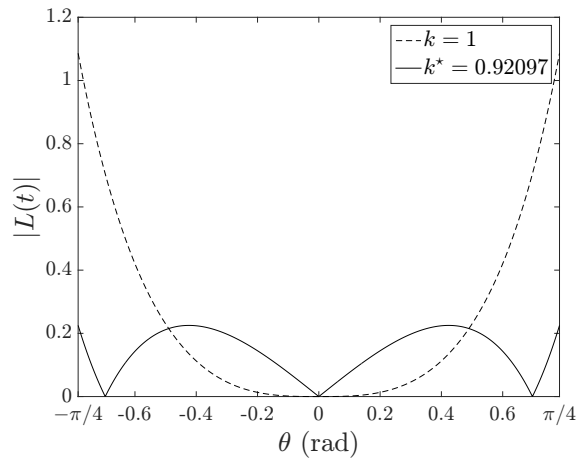

Fig. 4. Magnitude comparison between the optimal linearization law ($k^\star = 0.92097$) and the non-optimal one ($k = 1$) with $\bar{\theta} = \pi/4$.

By using the sum law, multiplication law, and the composite function law of the limit computation [27, p. 35] on (20), it can be obtained that

$$\cos z + z \sin z = \pi/2, \tag{21}$$

where $z = \lim_{\bar{\theta}\to(\pi/2)^-} \theta^\star$. It can be easily seen that (21) has only one solution, that is $z = \pi/2$. Therefore, $\lim_{\bar{\theta}\to(\pi/2)^-} \theta^\star = \pi/2$, which proves the property 1).

For the proof of the property 2), we can take the limit as $\bar{\theta} \to (\pi/2)^-$ for both sides of (18), which yields

$$\lim_{\bar{\theta}\to(\pi/2)^-} k^\star = \frac{1}{\cos z + z \sin z}. \tag{22}$$

By using the property 1), we have $\lim_{\bar{\theta}\to(\pi/2)^-} k^\star = 2/\pi$.

At the optimal solution, from (75) of Appendix A, we have

$$\max_{\theta\in[-\bar{\theta},\bar{\theta}]} |L^\star(t)| = g\sqrt{f_c(k^\star)} = g\sqrt{f_b(k^\star)}. \tag{23}$$

Taking the limit as $\bar{\theta} \to (\pi/2)^-$ on (74) (Appendix A) yields

$$\lim_{\bar{\theta}\to(\pi/2)^-} \sqrt{f_b(k^\star)} = \lim_{\bar{\theta}\to(\pi/2)^-} \frac{\sin\bar{\theta} - k^\star\bar{\theta}}{\cos\bar{\theta}}. \tag{24}$$

From the property 2), we have $\lim_{\bar{\theta}\to(\pi/2)^-} k^\star = 2/\pi$. Therefore, the limit in (24) is in a $0/0$ indeterminate form. Applying the L'Hôpital rule [27, p. 192] on (24) results in

$$\lim_{\bar{\theta}\to(\pi/2)^-} \sqrt{f_b(k^\star)} = \lim_{\bar{\theta}\to(\pi/2)^-} \frac{\cos\bar{\theta} - k^\star}{-\sin\bar{\theta}} = \frac{2}{\pi}. \tag{25}$$

Therefore, the result $\lim\limits_{\bar{\theta}\to(\pi/2)^-} \max\limits_{\theta\in[-\bar{\theta},\bar{\theta}]} |L^\star(t)| = 2g/\pi$ follows, which proves the property 3). □

Note that, the property 3) implies that the optimal linearization law is bounded for all swing angles $\theta \in (-\pi/2,\ \pi/2)$ by $2g/\pi$ if the gain $k^\star = 2/\pi$ is utilized. Let $L_1(t) := L(t)|_{k=1}$ be the conventional linearization law. It can be easily shown that when $\theta = \pm\bar{\theta}$, we have $\lim_{\bar{\theta}\to(\pi/2)^-} |L_1(t)| = +\infty$, which is unbounded. Therefore, the optimal linearization law is more favorable than the conventional one.

As an illustrative example, the optimization problem (17) will be solved by means of Theorem 1 with $\bar{\theta} = \pi/4$ rad. In

the first step, $\theta^\star$ can be computed from (19) by *fzero* routine of MATLAB. Then, $k^\star$ can be easily determined by (18). The resultant optimal solution is $k^\star = 0.92097$. According to Fig. 4, the maximum magnitude of the optimal linearization law is only 0.225, which is significantly smaller than that of the conventional linearization law with $k = 1$, whose maximum value is 1.086.

So far, the optimal value $k^\star$ of k can be calculated by (18) and (19). Substituting the control law (16) (with a replacement of k by $k^\star$) into (10), the closed loop system is governed by

$$l\ddot{\theta}(t) + k^\star g\theta(t) = -a\sin(\omega_0 t) - b\cos(\omega_0 t). \tag{26}$$

In order to generate the resonance, it must be chosen that $\omega_0 = \sqrt{k^\star g/l}$. Note that, (26) is a second order differential equation under the excitation of a biased sinusoidal input; thus, it has an analytical solution. The solution of (26) can be derived by using the method of undetermined coefficients [17]. A summation of the homogeneous solution and the particular solution of (26) gives us:

$$\begin{aligned}\theta(t) = {} & c_1\cos(\omega_0 t) + c_2\sin(\omega_0 t) \\ & + \frac{a}{2l\omega_0}t\cos(\omega_0 t) - \frac{b}{2l\omega_0}t\sin(\omega_0 t),\end{aligned} \tag{27}$$

where c_1 and c_2 are to-be-determined coefficients. By taking the first derivative of $\theta(t)$ in (27), it is obtained that

$$\begin{aligned}\dot{\theta}(t) = {} & -c_1\omega_0\sin(\omega_0 t) + c_2\omega_0\cos(\omega_0 t) + \frac{a}{2l\omega_0}\cos(\omega_0 t) \\ & - \frac{a}{2l}t\sin(\omega_0 t) - \frac{b}{2l\omega_0}\sin(\omega_0 t) - \frac{b}{2l}t\cos(\omega_0 t).\end{aligned} \tag{28}$$

The initial conditions $\theta(0) = 0$ and $\dot{\theta}(0) = 0$ are respectively applied on (27) and (28) to solve for two unknowns c_1 and c_2. Consequently, we have $c_1 = 0$ and $c_2 = -a/\left(2l\omega_0^2\right)$. In summary, the solution of (26) can be expressed by

$$\begin{aligned}\theta(t) &= -\frac{a}{2l\omega_0^2}\sin(\omega_0 t) + \frac{a}{2l\omega_0}t\cos(\omega_0 t) - \frac{b}{2l\omega_0}t\sin(\omega_0 t) \\ \dot{\theta}(t) &= -\frac{b}{2l\omega_0}\sin(\omega_0 t) - \frac{a}{2l}t\sin(\omega_0 t) - \frac{b}{2l}t\cos(\omega_0 t).\end{aligned} \tag{29}$$

By summing three sinusoidal functions with an identical frequency, $\theta(t)$ can be rewritten in a more compact form:

$$\theta(t) = \Gamma(t)\sin(\omega_0 t + \Phi(t)). \tag{30}$$

In (30), $\Gamma(t)$ and $\Phi(t)$ are given by

$$\begin{aligned}\Gamma(t) &= \sqrt{\left(\frac{\gamma t}{2l\omega_0}\right)^2 + \left(\frac{a}{2l\omega_0^2}\right)^2 - \frac{\gamma a t\cos\varphi}{2l^2\omega_0^3}} \\ \Phi(t) &= \tan^{-1}\left[\frac{(\gamma t\sin\varphi)/(2l\omega_0)}{-a/(2l\omega_0^2) + (\gamma t\cos\varphi)/(2l\omega_0)}\right],\end{aligned}$$

where $\gamma = \sqrt{a^2+b^2}$ and $\varphi = \tan^{-1}(-a/b)$. From (30), it is recognized that amplitude of the oscillation is $\Gamma(t)$. Furthermore, if $\gamma \neq 0$, we have

$$\begin{aligned}\lim_{t\to+\infty}\Gamma(t) &= \lim_{t\to+\infty}\sqrt{\left(\frac{\gamma t}{2l\omega_0}\right)^2 + \left(\frac{a}{2l\omega_0^2}\right)^2 - \frac{\gamma a t\cos\varphi}{2l^2\omega_0^3}} \\ &= +\infty.\end{aligned} \tag{31}$$

The result in (31) implies that the amplitude of $\theta(t)$ will approach to infinity as $t \to +\infty$, which clearly indicates the resonance phenomenon. In the following subsection, all of the unknowns a, b, $\bar{\theta}$, $\theta^\star$, and t_f will be determined to fulfill the tossing control requirements.

B. Embedding the tossing control requirements

In order to solve for five unknown parameters a, b, $\bar{\theta}$, $\theta^\star$, and t_f, the tossing control requirements (15) need to be employed. For the sake of clear presentation, denote $h(a,b,\bar{\theta},\theta^\star,t) := \ddot{x}(t)$, where $\ddot{x}(t)$ is given by (16) with a replacement of k by $k^\star$. With the initial conditions $x(0) = \dot{x}(0) = 0$ (see Section II), it can be easily obtained that

$$\dot{x}(t) = \int_0^t h(a,b,\bar{\theta},\theta^\star,\sigma)d\sigma \tag{32}$$

$$x(t) = \int_0^t\int_0^\sigma h(a,b,\bar{\theta},\theta^\star,\rho)d\rho d\sigma. \tag{33}$$

The double integration in (33) can be reduced to a single one by using the Cauchy' formula for repeated integral [29, p. 193], which results in

$$x(t) = \int_0^t (t-\sigma)h(a,b,\bar{\theta},\theta^\star,\sigma)d\sigma. \tag{34}$$

From (32), the condition on zero trolley speed at t_f—$\dot{x}(t_f) = 0$—can be rewritten by

$$\int_0^{t_f} h(a,b,\bar{\theta},\theta^\star,t)dt = 0. \tag{35}$$

By using (34), the condition on the target payload position at t_f—$x(t_f) - l\sin\theta(t_f) = \bar{x}_p$—is exploited as

$$-l\sin\theta(t_f) + \int_0^{t_f}(t_f - t)h(a,b,\bar{\theta},\theta^\star,t)dt = \bar{x}_p. \tag{36}$$

The zero swing rate condition $\dot{\theta}(t_f) = 0$ can be developed by using (29), whose result is

$$-\frac{b}{2l\omega_0}\sin(\omega_0 t_f) - \frac{a}{2l}t_f\sin(\omega_0 t_f) - \frac{b}{2l}t_f\cos(\omega_0 t_f) = 0. \tag{37}$$

Now, we have four equations (19), (35)–(37) to solve for five unknowns. Thus, one missing equation can be filled in by setting $\theta(t_f) = -\bar{\theta}$. By using (29), it can be rewritten as

$$\frac{-a}{2l\omega_0^2}\sin(\omega_0 t_f) + \frac{t_f}{2l\omega_0}\left[a\cos(\omega_0 t_f) - b\sin(\omega_0 t_f)\right] = -\bar{\theta}. \tag{38}$$

In summary, to solve for five unknowns a, b, $\bar{\theta}$, $\theta^\star$, and t_f, we have the following set of five nonlinear equations

$$\begin{cases}\dfrac{-a}{2l\omega_0^2}\sin(\omega_0 t_f) + \dfrac{t_f}{2l\omega_0}\left[a\cos(\omega_0 t_f) - b\sin(\omega_0 t_f)\right] = -\bar{\theta} \\ -\dfrac{b}{\omega_0}\sin(\omega_0 t_f) - at_f\sin(\omega_0 t_f) - bt_f\cos(\omega_0 t_f) = 0 \\ \displaystyle\int_0^{t_f} h(a,b,\bar{\theta},\theta^\star,t)dt = 0 \\ l\sin\bar{\theta} - \displaystyle\int_0^{t_f} h(a,b,\bar{\theta},\theta^\star,t)tdt = \bar{x}_p \\ \sin\left(\bar{\theta}+\theta^\star\right) - \theta^\star\cos\left(\bar{\theta}+\theta^\star\right) = \bar{\theta}.\end{cases} \tag{39}$$

The MATLAB routine *fsolve* can be utilized to solve (39). Initial guesses of unknowns (for feeding to the *fsolve* solver) are $a_0 = 0$, $b_0 = 0$, $\bar{\theta}_0 = \pi/12$, $\theta_0^\star = \bar{\theta}_0/2$, and $t_{f0} = 2n\pi/\omega$ $(n = 1, 2, 3, \ldots)$ where $\omega = \sqrt{g/l}$. Note that, in order to ensure the maximum velocity and maximum acceleration constraints are strictly complied, namely $\max|\dot{x}(t)| \leq v_{\max}$ and $\max|\ddot{x}(t)| \leq a_{\max}$, the index n of t_{f0} can be gradually increased until those inequalities are met.

Remark 1. Since $\bar{\theta}$, $\theta^\star$, and t_f are required to be positive, hence, in (39), the following changes of variables should be performed to ensure *fsolve* routine run properly: $\bar{\theta} = z_1^2$, $\theta^\star = z_2^2$, and $t_f = z_3^2$. Moreover, for safety, the payload should remain underneath the trolley, namely that magnitude of the terminal swing angle should not exceed $\pi/2$. In other words, it requires that $\bar{\theta} < \pi/2$. To guarantee such a condition for the resultant solution, the following auxiliary barrier equation is added to (39):

$$z_4 := \ln\left(\pi/2 - \bar{\theta}\right) = \ln\left(\pi/2 - z_1^2\right). \tag{40}$$

Therefore, at present, six unknowns need to be solved are a, b, z_i $(i = 1, \ldots, 4)$. Initial guesses for z_i $(i = 1, \ldots, 4)$ are made at $z_{10} = \sqrt{\bar{\theta}_0}$, $z_{20} = \sqrt{\theta_0^\star}$, $z_{30} = \sqrt{t_{f0}}$, and $z_{40} = \ln\left(\pi/2 - \bar{\theta}_0\right)$.

It is also worth noting that the resonance-based tossing controller only utilizes the rope length l in its design, which can be easily measured in practice. Therefore, parametric uncertainty is not a significant issue for the proposed method.

C. Transferring and returning phases

An entire cycle of the transportation is explained as follows. In the first half (or transferring phase), by employing the control law (16), the payload is transferred from the initial position to the discharging destination, at which the grab opens to drop down the materials. The second half (or returning phase) starts immediately after the first half, where the trolley returns to the initial position to supply materials for the next cycle. Note that, at the end of the second half, the trolley speed must be zero and there should be no residual vibration. Since the controller (16) is independent of the payload mass m, one simple way to realize the above requirement is to apply $\ddot{x}(t)$ (in (16)) backward in time as the control input for the returning phase. The proof of this claim is provided as follows. Recall the dynamical model of the overhead crane in the transferring phase as below (see (10)):

$$l\ddot{\theta}(t) + g\sin\theta(t) = \cos\theta(t)\ddot{x}(t),\ t \in [0,\ t_f], \tag{41}$$

where, by assumption, the system starts from rest (see Section II). Namely, the initial conditions of (41) are

$$x(0) = \dot{x}(0) = \theta(0) = \dot{\theta}(0) = 0. \tag{42}$$

Denote $\bar{x}$ as the terminal position of the trolley at the end of the transferring phase, that is to say, $\bar{x} = \bar{x}_p - l\sin\bar{\theta}$. Since the transferring phase employs the resonance-based tossing controller, the following terminal conditions of (41) are realized:

$$x(t_f) = \bar{x},\ \dot{x}(t_f) = 0,\ \theta(t_f) = -\bar{\theta},\ \dot{\theta}(t_f) = 0. \tag{43}$$

Now consider the model of the overhead crane system in the returning phase as below:

$$l\ddot{\theta}_r(\mu) + g\sin\theta_r(\mu) = \cos\theta_r(\mu)\ddot{x}_r(\mu),\ \mu \in [t_f,\ 2t_f]. \tag{44}$$

In (44), to draw a distinction with the transferring phase, the swing angle and the control input of the returning phase are denoted as θ_r and $\ddot{x}_r$ respectively. Since the returning phase starts immediately after the transferring phase, to realize the differentiable continuity condition of the solution, it requires that $\theta_r(t_f) = \theta(t_f)$, $\dot{\theta}_r(t_f) = \dot{\theta}(t_f)$, $x_r(t_f) = x(t_f)$, and $\dot{x}_r(t_f) = \dot{x}(t_f)$, namely that the initial conditions of (44) are assumed to be

$$x_r(t_f) = \bar{x},\ \dot{x}_r(t_f) = 0,\ \theta_r(t_f) = -\bar{\theta},\ \dot{\theta}_r(t_f) = 0. \tag{45}$$

The assumption of (45) will be checked at the end of the proof. Since the control input $\ddot{x}_r(\mu)$ is a backward-in-time version of the input $\ddot{x}(t)$, the following relation holds:

$$\ddot{x}_r(\mu) = \ddot{x}(2t_f - \mu),\ \forall\mu \in [t_f,\ 2t_f]. \tag{46}$$

By substituting (46) to (44), we have

$$l\ddot{\theta}_r(\mu) + g\sin\theta_r(\mu) = \cos\theta_r(\mu)\ddot{x}(2t_f - \mu),\ \mu \in [t_f,\ 2t_f]. \tag{47}$$

Performing the following time-reversing transformation without shifting/scaling the magnitude of the variable θ_r:

$$\tau = 2t_f - \mu,\ \psi = \theta_r. \tag{48}$$

From (48), the following results can be obtained:

$$d\tau = -d\mu,\ d\tau^2 = d\mu^2,\ d\psi = d\theta_r. \tag{49}$$

Applying the transformations of (48) and (49) on (47) results in the following differential equation:

$$l\ddot{\psi}(\tau) + g\sin\psi(\tau) = \cos\psi(\tau)\ddot{x}(\tau),\ \tau \in [0,\ t_f], \tag{50}$$

with the subsequent terminal conditions

$$x(t_f) = \bar{x},\ \dot{x}(t_f) = 0,\ \psi(t_f) = -\bar{\theta},\ \dot{\psi}(t_f) = 0. \tag{51}$$

After the transformation, the differential equation (50) (describing the returning phase dynamic) is identical with (41), which represents the dynamic of the transferring phase. Furthermore, their terminal conditions are also similar (see (43) and (51)). Hence, the following conclusion can be made:

$$\psi(\tau) = \theta(\tau),\ \dot{\psi}(\tau) = \dot{\theta}(\tau),\ \forall\tau \in [0,\ t_f]. \tag{52}$$

From (48), we have $\psi = \theta_r$ and by using $d\tau = -d\mu$ from (49), the result of (52) gives us

$$\theta_r(\mu) = \theta(\tau),\ \dot{\theta}_r(\mu) = -\dot{\theta}(\tau),\ \forall\tau \in [0,\ t_f]. \tag{53}$$

By employing the relation between two time frames $\tau = 2t_f - \mu$ on (53), we have

$$\theta_r(2t_f) = \theta(0) = 0,\ \dot{\theta}_r(2t_f) = -\dot{\theta}(0) = 0. \tag{54}$$

Furthermore, from (46), we have $\ddot{x}_r(\mu) = \ddot{x}(\tau)$, therefore

$$x_r(\mu) = x(\tau),\ \dot{x}_r(\mu) = -\dot{x}(\tau),\ \forall\tau \in [0,\ t_f]. \tag{55}$$

By using the relation $\tau = 2t_f - \mu$, (55) gives us

$$x_r(2t_f) = x(0) = 0,\ \dot{x}_r(2t_f) = -\dot{x}(0) = 0. \tag{56}$$

The results in (54) and (56) show that, at the end of the returning phase, the trolley returns to the initial position at zero speed and no residual vibration occurs. In order to finish the proof, the assumption on the differentiable continuity of the solution must be checked. From (53) and (55), by utilizing $\tau = 2t_f - \mu$, the following can be obtained:

$$x_r(t_f) = \bar{x},\ \dot{x}_r(t_f) = 0,\ \theta_r(t_f) = -\bar{\theta},\ \dot{\theta}_r(t_f) = 0, \tag{57}$$

which verifies (45). Consequently, the continuity assumption is satisfied. This completes the proof.

Remark 2. The tossing control requirement (15) implies that, at the discharging destination, only when all materials can be fully discharged from the grab bucket *instantaneously*, the positioning accuracy of the falling materials is perfectly ensured. Given that a 4-ton capacity grab bucket needs two seconds to drop its entire load, an absolute positioning accuracy may not be possible in reality. However, when the tossing control is applied, it can be guaranteed that the velocities of the payload (i.e. the grab bucket with materials inside) in both horizontal and vertical directions are zero at the discharging destination (see (15)). Consequently, in the neighborhood of the dropping point, the payload's velocities are sufficiently small for a discharge with an acceptable accuracy. Furthermore, the contact forces between particles of the granular materials as well as between particles and the inner wall of the grab bucket will make the stream of the dropping materials nearly vertical, despite having slight initial velocities. According to the above discussions, therefore, the grab can prematurely open one second (half of the grab's fully discharging time) before reaching the discharging destination, by which a satisfactory positioning accuracy is ensured. By using such a discharging operation, we can effectively prevent the falling materials spilling over the container without slowing down the transportation. The above discharging process can be observed in the operation of experienced crane operators. In summary, the discharging period is two seconds, whose center is the terminal time t_f obtained by solving (39).

IV. Simulation Result: Comparison with Minimum-Time Vibration Control

A real gantry crane in use at the harbor is utilized in the simulation study. It has a capacity of $\sim 65\%$ compared with the Sepetiba crane introduced in [24]. The maximum velocity and maximum acceleration are $v_{\max} = 1.5$ m/s and $a_{\max} = 0.6$ m/s^2. The rope length is $l = 20$ m. The resonance-based tossing control will be compared with the minimum-time vibration control in various discharging destination $\bar{x}_p$ ranging from 9 m to 20 m. Therefore, a discrete-value simulation test is configured as follows. Let Σ be the range of the discharging distance $\bar{x}_p$ as below:

$$\Sigma := \left\{\bar{x}_p \in \mathbb{R}^+ : d \le \bar{x}_p \le D, d = 9 \text{ m}, D = 20 \text{ m}\right\}, \tag{58}$$

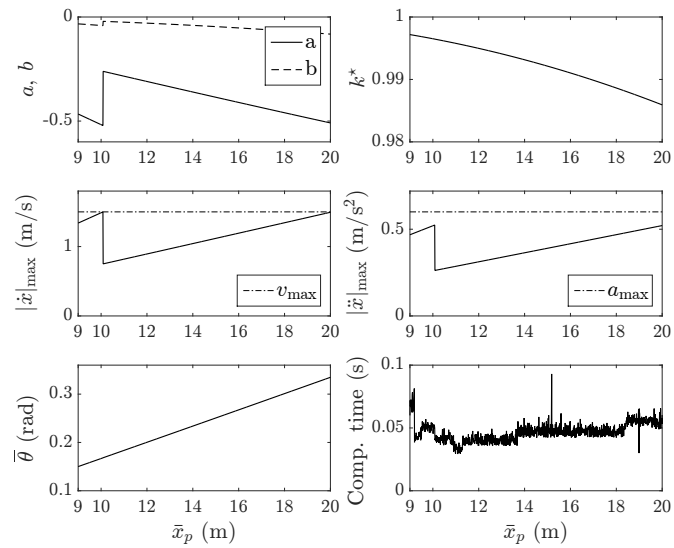

Fig. 5. Computational result of the resonance tossing control in the discrete-value simulation test.

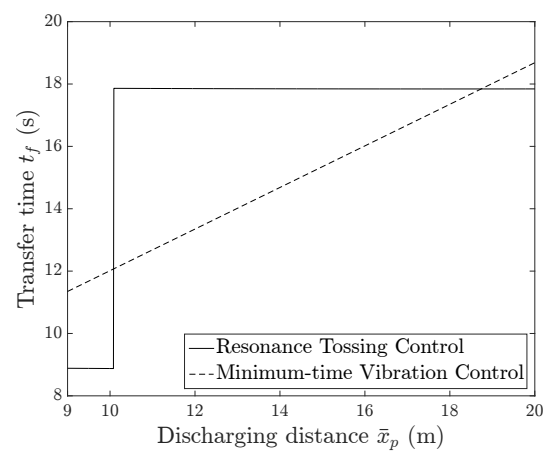

Fig. 6. Comparison of the transferring time between the resonance-based tossing and the minimum-time vibration controls in the discrete-value simulation test.

where $\mathbb{R}^+$ is the set of real numbers. The set Σ is uniformly discretized into $N = 1100$ points to form the following subset $\Sigma_D \subset \Sigma$:

$$\Sigma_D = \left\{\bar{x}_p \in \Sigma : \bar{x}_p = d + i\frac{D-d}{N}, i \in \mathbb{N}, 0 \le i \le N\right\}. \tag{59}$$

Note that $\mathbb{N}$ is the natural numbers set. For each point of Σ_D, the solution of the concerning (tossing or vibration) controller will be solved by using the results in Section III-B and Appendix B respectively. The controller gives a positive result to the test if a solution exists for very member of Σ_D and satisfies the velocity constraint $|\dot{x}(t)| \le v_{\max}$ and the acceleration constraint $|\ddot{x}(t)| \le a_{\max}$ simultaneously.

Firstly, the test will be applied to the resonance tossing control. The result is positive. The control parameters as well as the computational results of the test are shown in Fig. 5. It can be seen that the magnitude of the terminal swing angle $\bar{\theta}$ increases as the transferring distance increases, which agrees

TABLE I
COMPUTATIONAL RESULTS OF THE RESONANCE-BASED TOSSING CONTROL AND THE MINIMUM-TIME VIBRATION CONTROL.

Proposed resonance-based tossing control ($\bar{x}_p$ = 10 m)							Minimum-time vibration control ($\bar{x}_p$ = 10 m)						
a	b	$\bar{\theta}$	$\theta^\star$	$k^\star$	t_f	$2t_f$	λ	ζ	η	$t_1^\star$	$t_2^\star$	$t_f^\star$	$2t_f^\star$
-0.517	-0.041	0.167 rad	0.084 rad	0.997	**8.86 s**	**17.72 s**	1	1.373	0.996	5.344 s	6.671 s	**12.01 s**	**24.02 s**

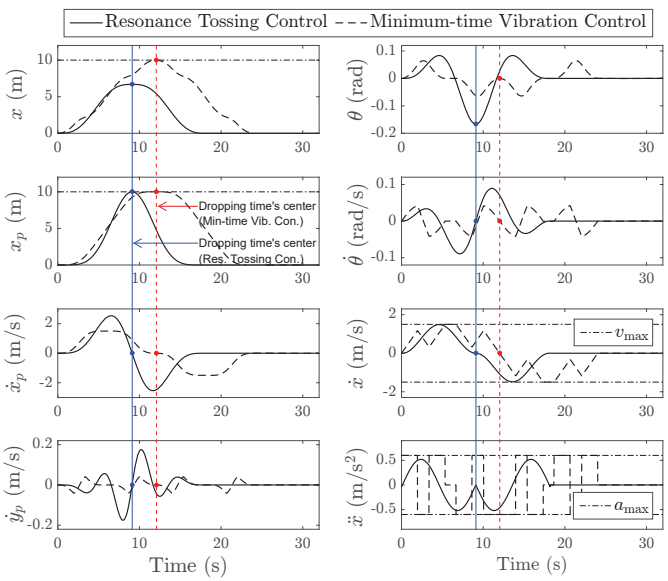

Fig. 7. Comparative simulation results of the proposed resonance-based tossing control and the minimum-time vibration control.

well with the physical intuition. Furthermore, $\bar{\theta}$ is always less than $\pi/2$, thus the safety condition is ensured. The maximum and average computational time are 0.093 second and 0.0467 second respectively on a personal computer with an Intel Core i5-3210M and 8 GB of RAM. Therefore, the applicability of the resonance tossing controller is verified.

The discrete-value simulation test is then carried out on the minimum-time vibration control. The transferring time comparison of the resonance tossing control and the minimum-time vibration control are shown in Fig. 6. According to Fig. 6, the resonance tossing control can provide a significantly faster transferring time than the minimum-time vibration control when $\bar{x}_p \in [9,\ 10.08]$ m, namely in configurations with short discharging destination. At its peak, for the discharging distance $\bar{x}_p = 10.08$ m, the transferring time of the resonance tossing control is 8.87 seconds, whereas the minimum-time vibration control needs 12.07 seconds. Therefore, by using the tossing control method, one can reduce the transferring time by 3.2 seconds (or 6.4 seconds in one transportation cycle). Thus, productivity can be enhanced up to 26.5%. This evidence confirms the main purpose of this paper, and proves the possibility of breaking the time limitation of the minimum-time vibration control. For the long transferring distance contexts, the resonance tossing control is not able to produce a similar result. Here, the characteristic of the proposed tossing controller, which is inherited from the resonance behavior, can be clearly observed, namely that its transferring time lies in the neighborhood of $T, 2T, \ldots,$ where T is the natural frequency of the closed-loop system.

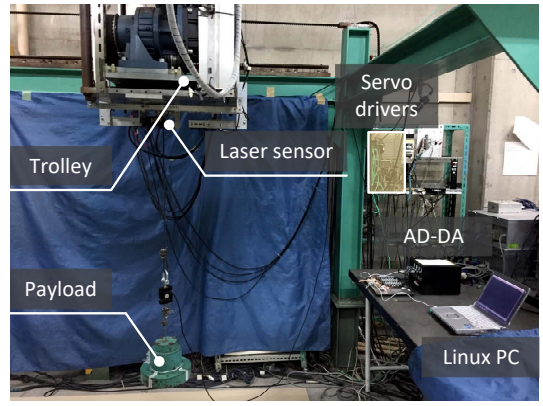

Fig. 8. Structure of the experimental overhead crane system.

In order to demonstrate a whole transportation process including both transferring and returning phases, two controllers are computed for the desired discharging distance $\bar{x}_p = 10$ m. The resultant control parameters of two controllers are aggregated in Table I. Simulation results of two controllers are shown in Fig. 7. It can be seen that, at the center of the materials dropping/discharging period (represented by the solid blue line and the dash red line for the resonance-based tossing control and the minimum-time vibration control respectively), the payload reaches the target destination at zero horizontal and vertical velocities, whereas the trolley completely stops. Therefore, requirements of the transferring phase are fulfilled by both controllers. Moreover, at the end of the returning phase, in both control schemes, the vibration is completely suppressed. To sum up, two controllers satisfy the necessary conditions of the bulk materials transportation. The main difference between the proposed resonance-based tossing control and the minimum-time vibration control is explained as follows. At the dropping time, the swing angle in the case of tossing controller is $-\bar{\theta} = -0.167$ rad (or -9.57 degrees), whereas it is 0 degree when the minimum-time vibration control is employed (see θ graph in Fig. 7). This characteristic leads to the fact that, at the discharging point, the payload must not be exactly under the trolley in the tossing control case, which is completely in contrast to the minimum-time vibration control (see x and x_p graphs of Fig. 7). For this reason, if the actuator limits are sufficiently large (or the discharging distance is short), the tossing controller can take full advantage of the long rope length to save the transferring time since the trolley is not required to travel a full distance as it does in the case of vibration controller (see subfigure x).

V. EXPERIMENTAL RESULT

The proposed resonance-based tossing control now will be implemented on a small-scale overhead crane system,

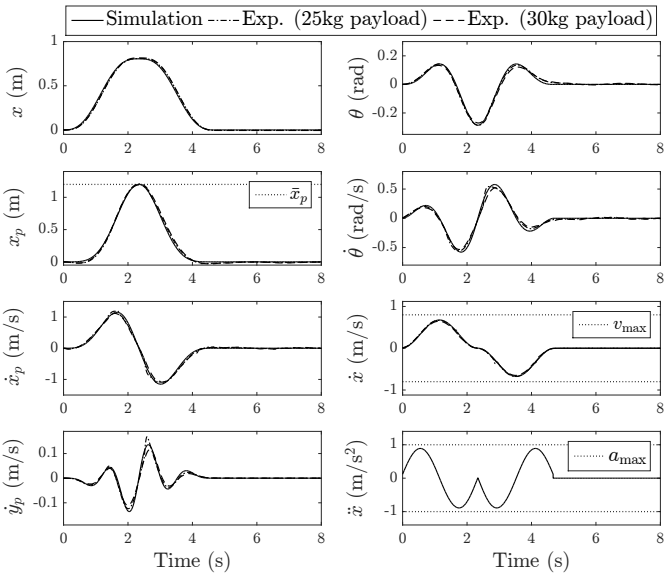

Fig. 9. Experiment (Exp.) results of the resonance tossing control. They are in a complete agreement with the simulation result.

which is illustrated in Fig. 8. The actuator is a YASKAWA USAFED-09C2L1K AC servo motor and the servo driver is a MITSUBISHI MR-J2S-200A, which are set to the position control mode. MATLAB/Simulink is utilized to compose the program and build the executing file via Real Time Workshop. A Fedora Linux PC operates the executing file in real time with a sampling period of 10 ms via the Real Time Application Interface (RTAI). Moreover, a pair of laser sensors KEYENCE VG-035 is used to measure the swing angle. The rope length is $l = 1.4$ m. Based on specifications of the servo motor and the transmission system, values of the maximum velocity and the maximum acceleration are given by $v_{\max} = 0.8$ m/s and $a_{\max} = 1$ m/s^2 respectively. The target horizontal position of the payload is $\bar{x}_p = 1.2$ m. According to the result of Section III, parameters of the resonance tossing control for the experimental configuration can be computed as $a = -0.8763$, $b = -0.1187$, $\bar{\theta} = 0.2861$ rad, $\theta^\star = 0.1444$ rad, $k^\star = 0.9897$, and $t_f = 2.336$ seconds. The computational time is only 0.14 second; thus, online command generation is feasible. Note that, the resonance-based tossing controller is realized as a feedforward scheme, namely that it is offline computed and integrated twice to obtain a control law in the position level, which is equivalent to (34). This position command will be realized by a built-in motion controller of the servo driver to drive the trolley motion. For clarification, the measured swing angle is recorded for a sole purpose of verifying the validity of the proposed controller in practice.

To illustrate the independence (or robustness) of the proposed resonance tossing control law with respect to the payload mass, two experiments with a 25 kg and a 30 kg payload are conducted. In both cases, the same controller parameters computed as above are employed. Fig. 9 depicts the experimental results of the resonance tossing controller. The system responses in two cases (25 kg and 30 kg payload) are almost identical, which indicates that the proposed controller is robust against the change of the payload mass. The payload is precisely transferred to the destination with zero horizontal and vertical velocities; afterward, it is brought back to the initial position with no residual vibration. Note that, the noise level of the experimental acceleration signal, which is obtained by differentiating the measured position (from an encoder attached at the end of the servo motor) twice, is very high even though a low-pass filter is employed, hence, it is not present in Fig. 9. Such an observation noise has no effect to the system performance. The experimental data completely agrees with the simulation result, which verifies the effectiveness of the proposed resonance-based tossing control system.

VI. Discussion on possible future work

In this paper, it has been shown that, compared with the classical minimum-time vibration control, it is possible to reduce the transferring time for bulk materials transportation by using tossing control methodology. This result, therefore, indicates the importance of further research on the topic. This paper introduced a specific type of tossing controller, which is based on the phenomenon of mechanical resonance, to realize transportation requirements. Nevertheless, in some configurations, transferring time using the resonance-based tossing controller is slower than the minimum-time vibration control. Note that, the objective of both tossing and vibration controls is to reduce the transferring time as much as possible. As a result, the use of a particular controller will vary according to the circumstance. In order to address this issue, a *unified minimum-time controller*, which satisfies the requirements of the bulk materials transferring process (stated in Section I) in minimum time, should be formulated in future. Actuator limits, including maximum velocity and maximum acceleration, must be taken into account. The Pontryagin Minimum Principle is the main tool to design such controller. There may have two important points. First, the terminal swing angle $\theta(t_f)$ should be free instead of constrained by $\theta(t_f) = 0$ as in the case of the vibration control. If the time-optimal solution requires that $\theta(t_f) \neq 0$ then the unified minimum-time control is actually a minimum-time tossing controller; otherwise, it is a minimum-time vibration controller. By that means, it is guaranteed that the unified minimum-time controller is the best choice among the tossing and vibration control groups. Second, since a large swing angle may occur, a nonlinear model must be utilized, and therefore an analytical solution of the unified minimum-time control might not be possible. As a result, a computational nonlinear optimal control scheme may be needed.

Additionally, in order to increase the maneuverability, time-varying rope length and a unified minimum-time control design for three-dimensional crane systems should be considered.

Lastly, the resonance-based tossing control proposed in this paper is indeed a feedforward scheme. Thus, it cannot compensate for harsh external disturbances such as strong wind or sea waves if it is utilized in outdoor environments. Furthermore, an input deadzone may exist, which might lessen the control performance. In order to handle such issues, intelligent schemes such as neural network [30], [31] can be incorporated to equip the tossing control with a disturbance rejection capability to enhance the overall control performance.

VII. Conclusions

This paper proposed a resonance-based tossing controller for an overhead crane to suit for the context of bulk materials transportation. The analysis showed that both tossing control and vibration control groups can realize the requirements of the bulk materials transferring process. The crucial conclusion of the paper emphasizes on the result that there are possibilities to break the time limitation of the minimum-time vibration controller—the fastest member of the vibration control family—by using the tossing control methodology. Both simulation and experimental results are provided to clarify the effectiveness of the proposed method and they are in a complete agreement. Possible developments are elaborated to further enhance productivity of the bulk materials transportation in future.

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Appendix A
Proof of Theorem 1

Since $f(k,\theta)$ is an even function with respect to θ, hence it is sufficient to perform the optimization procedure on $\theta \in \left[0,\bar{\theta}\right]$. Flow of the proof is explained as follows. The interest region of k is divided into three sub-intervals, namely $k \in (0,+\infty) = (0,\underline{k}] \cup [\underline{k},1] \cup [1,+\infty)$, where $\underline{k} := 1/\left(\cos\bar{\theta}+\bar{\theta}\sin\bar{\theta}\right)$. Firstly, it is proven that for $0 < k \leq \underline{k}$, the minimum value of the function $\max f(k,\theta)$ attains at $k=\underline{k}$. Secondly, for $1 \leq k < +\infty$, the minimum of $\max f(k,\theta)$ is achieved at $k=1$. Therefore, the whole interest region of k shrinks from $(0,+\infty)$ down to $[\underline{k},1]$. Finally, for $\underline{k} \leq k \leq 1$, the optimal solution is determined by the intersection between the critical function and the boundary function.

To begin the proof, following essential calculus results are provided to facilitate the subsequent presentations

$$\theta - \sin\theta \geq 0,\ \forall\theta \geq 0. \tag{60}$$

$$\sin\theta - \theta\cos\theta \geq 0,\ \forall\theta \geq 0. \tag{61}$$

Furthermore, the first partial derivative of $f(k,\theta)$ with respect to k is given by

$$\frac{\partial f(k,\theta)}{\partial k} = \frac{2\theta\left(k\theta-\sin\theta\right)}{\cos^2\theta}. \tag{62}$$

A. Interval 1 ($0 < k \leq \underline{k}$)

At first, let us consider a function $q(\theta) = \underline{k}\theta - \sin\theta$. The boundary values of $q(\theta)$ are given by

$$q(\theta)|_{\theta=0} = 0 \tag{63}$$

$$q(\theta)|_{\theta=\bar{\theta}} = \frac{\cos\bar{\theta}\left(\bar{\theta}\cos\bar{\theta}-\sin\bar{\theta}\right)}{\cos\bar{\theta}+\bar{\theta}\sin\bar{\theta}}. \tag{64}$$

By using the result of (61) and note that $\bar{\theta} \geq 0$, we have $q(\theta)|_{\theta=\bar{\theta}} \leq 0$. In summary, we have

$$q(\theta)|_{\theta=0} = 0 \text{ and } q(\theta)|_{\theta=\bar{\theta}} \leq 0. \tag{65}$$

The critical point θ_s of $q(\theta)$ must satisfy $\partial q(\theta)/\partial\theta|_{\theta=\theta_s} = 0 \Rightarrow \cos\theta_s = \underline{k}$. Therefore, the value of $q(\theta)$ at the critical point θ_s is given by

$$q(\theta_s) = \underline{k}\theta_s - \sin\theta_s = \theta_s\cos\theta_s - \sin\theta_s. \tag{66}$$

By using (61), it is concluded that $q(\theta_s) \leq 0$. By combining this result and (65), we have $q(\theta) \leq 0,\ \forall\theta \in \left[0,\ \bar{\theta}\right]$. Therefore, $\underline{k}\theta - \sin\theta \leq 0,\ \forall\theta \in \left[0,\ \bar{\theta}\right]$. Moreover, in this case, we have $k \leq \underline{k}$; thus, $k\theta - \sin\theta \leq 0$. For this reason, from (62), $\partial f(k,\theta)/\partial k \leq 0$ holds. This implies that when θ is held constant, $f(k,\theta)$ decreases as k increases. Therefore

$$\min_k \max_{\theta\in\left[0,\ \bar{\theta}\right]} f(k,\theta)\Big|_{0<k\leq\underline{k}} = \max_{\theta\in\left[0,\ \bar{\theta}\right]} f(k,\theta)\Big|_{k=\underline{k}}. \tag{67}$$

B. Interval 2 ($1 \leq k < +\infty$)

From (60), we have $\sin\theta \leq \theta$, but $k \geq 1$, so $\sin\theta \leq k\theta \Rightarrow k\theta - \sin\theta \geq 0$. By using the condition that $\theta \geq 0$ and $\cos^2\theta > 0$, it can be obtained from (62) that $\partial f(k,\theta)/\partial k \geq 0$. This result means that when θ is held constant, $f(k,\theta)$ increases as k increases. Therefore, it can be easily seen that

$$\min_k \max_{\theta\in\left[0,\ \bar{\theta}\right]} f(k,\theta)\Big|_{k\geq 1} = \max_{\theta\in\left[0,\ \bar{\theta}\right]} f(k,\theta)\Big|_{k=1}. \tag{68}$$

C. Interval 3 ($\underline{k} \leq k \leq 1$)

Since the absolute maximum value of a function is the largest term between its values at the critical points and at the boundaries [27, p. 207]; thus, the critical function and the boundary function should be computed.

The critical function is derived as follows. For each k, a critical point θ_c of $f(k,\theta)$ must satisfy $\partial f(k,\theta)/\partial\theta|_{\theta=\theta_c} = 0$, which can be expanded to

$$\left(\sin\theta_c - k\theta_c\right)\cos\theta_c\left[1 - k\left(\cos\theta_c + \theta_c\sin\theta_c\right)\right] = 0. \tag{69}$$

The results of (69) are

$$\left[\begin{array}{l} \sin\theta_c = k\theta_c \\ \cos\theta_c = 0 \Rightarrow \theta_c = \pi/2 \Rightarrow \text{(Reject)} \\ 1 - k\left(\cos\theta_c + \theta_c\sin\theta_c\right) = 0. \end{array}\right. \tag{70}$$

In (70), when $\sin\theta_c = k\theta_c$, we have $f(k,\theta)|_{\theta=\theta_c} = 0$. Thus, this critical point can be excluded from consideration since $f(k,\theta)$ is always non-negative. For the remaining scenario, namely $1 - k\left(\cos\theta_c + \theta_c\sin\theta_c\right) = 0$, we have $k = 1/\left(\cos\theta_c + \theta_c\sin\theta_c\right)$. Denote $f_c(k)$ as the value of $f(k,\theta)$ at the critical point θ_c. By using the result $k = 1/\left(\cos\theta_c + \theta_c\sin\theta_c\right)$, the critical function $f_c(k)$ can be computed by

$$f_c(k) = \left(\tan\theta_c - \frac{k\theta_c}{\cos\theta_c}\right)^2 = \left(\frac{\sin\theta_c - \theta_c\cos\theta_c}{\cos\theta_c + \theta_c\sin\theta_c}\right)^2. \tag{71}$$

The first derivatives of $f_c(k)$ and k with respect to θ_c are

$$\frac{df_c(k)}{d\theta_c} = \frac{2\theta_c^2\left(\sin\theta_c - \theta_c\cos\theta_c\right)}{\left(\cos\theta_c + \theta_c\sin\theta_c\right)^3}. \tag{72}$$

$$\frac{dk}{d\theta_c} = \frac{-\theta_c\cos\theta_c}{\left(\cos\theta_c + \theta_c\sin\theta_c\right)^2} \leq 0. \tag{73}$$

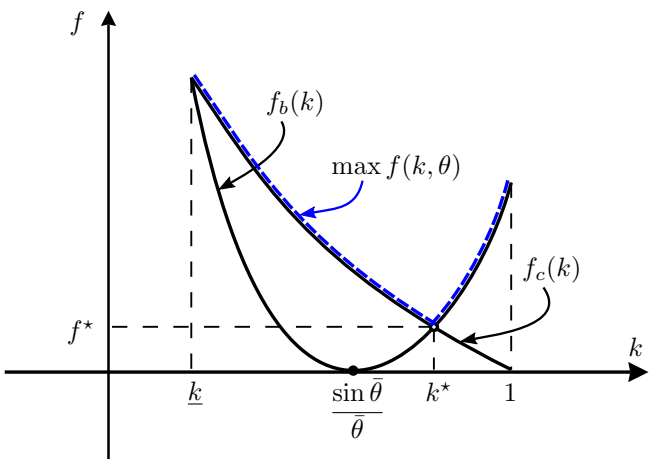

Fig. 10. Illustrations of the critical function $f_c(k)$, the boundary function $f_b(k)$, and the function $\max f(k,\theta)$. The optimal solution is attained at the second intersection between $f_c(k)$ and $f_b(k)$.

By using the result of (61), it can be concluded that $df_c(k)/d\theta_c \geq 0$. For this reason, $f_c(k)$ increases as θ_c increases. Furthermore, (73) implies that k decreases as θ_c increases. Therefore, in summary, we have the critical function $f_c(k)$ decreases with respect to k.

Now, we will examine the boundary function. Since $f(k,\theta)|_{\theta=0} = 0$, the only boundary needing to be considered is $\theta = \bar{\theta}$. Denote $f_b(k)$ as the boundary function of $f(k,\theta)$. It can be evaluated as

$$f_b(k) = f(k,\theta)|_{\theta=\bar{\theta}} = \left(\frac{\sin\bar{\theta} - k\bar{\theta}}{\cos\bar{\theta}}\right)^2. \tag{74}$$

From (74), $f_b(k)$ is a parabolic function with respect to k, which attains zero at $k = \left(\sin\bar{\theta}\right)/\bar{\theta}$.

At this point, properties of the critical function $f_c(k)$ and the boundary function $f_b(k)$ are clear. Their illustrations are depicted in Fig. 10. Note that $\max f(k,\theta) = \max\{f_b(k), f_c(k)\}$. Therefore, one could sketch the function $\max f(k,\theta)$ by the blue dash curve as shown in Fig. 10. The left part of the function $\max f(k,\theta)$ belongs to $f_c(k)$ and the right part belongs to $f_b(k)$. It is immediately recognized that $\max f(k,\theta)$ is a convex function. From that, it is not difficult to see that the minimizer $k^\star$ of the minimax optimization problem (17) is attained at the intersection between $f_c(k)$ and $f_b(k)$, namely $k^\star$ must satisfy

$$f_c(k^\star) = f_b(k^\star), \tag{75}$$

where $k^\star = 1/\left(\cos\theta_c^\star + \theta_c^\star \sin\theta_c^\star\right)$ and $\theta_c^\star$ is the critical point of $f(k^\star,\theta)$. By using (71) and (74), the condition in (75) can be expanded as follows

$$\left(\frac{\sin\theta_c^\star - \theta_c^\star\cos\theta_c^\star}{\cos\theta_c^\star + \theta_c^\star\sin\theta_c^\star}\right)^2 = \left(\frac{\sin\bar{\theta} - k^\star\bar{\theta}}{\cos\bar{\theta}}\right)^2. \tag{76}$$

The result in (76) can be further developed to

$$\left[\begin{array}{ll} \sin\left(\theta_c^\star - \bar{\theta}\right) - \theta_c^\star\cos\left(\theta_c^\star - \bar{\theta}\right) = -\bar{\theta} & (77) \\ \sin\left(\theta_c^\star + \bar{\theta}\right) - \theta_c^\star\cos\left(\theta_c^\star + \bar{\theta}\right) = \bar{\theta}. & (78) \end{array}\right.$$

It can be seen that (77) corresponds to the first intersection point (at $k = \underline{k}$) of $f_c(k)$ and $f_b(k)$ (see Fig. 10); hence, it does not indicate the optimal solution. The second intersection point given by (78) determines our desirable optimal solution. From this observation as well as the results of (67) and (68), the minimizer of the optimization problem is given by

$$k^\star = \frac{1}{\cos\theta^\star + \theta^\star\sin\theta^\star}, \tag{79}$$

where $\theta^\star$ is computed by $\sin\left(\bar{\theta} + \theta^\star\right) - \theta^\star\cos\left(\bar{\theta} + \theta^\star\right) = \bar{\theta}$. This completes the proof.

Appendix B
Review of the Minimum-Time Vibration Control

In similarity to the resonance-based tossing control, the minimum-time vibration control is designed only for the transferring phase. Control input for the returning phase can be obtained by simply applying the computed input of the transferring phase backward in time.

In the transferring phase, the objective of the minimum-time vibration control is to transfer the payload to the discharging destination without vibration (i.e. $x(t_f^\star) = \bar{x}_p$, $\dot{x}(t_f^\star) = \dot{\theta}(t_f^\star) = 0$, *and* $\theta(t_f^\star) = 0$) in a minimum time $t_f^\star$. Note that $\omega = \sqrt{g/l}$. For a sufficient large transferring distance, a version of the minimum-time vibration control profile under the maximum velocity and maximum acceleration constraints is illustrated in Fig. 11, where a bang-bang control input suggested by the Pontryagin Minimum Principle is observed. Specifically, the dimensionless durations ζ and η are computed by [22], [25], [26]

$$\begin{aligned} K_m &= 2\zeta - (2\lambda - 1)\eta + 2(\lambda - 1)\pi \\ \eta &= 2\tan^{-1}\left(\frac{\sin\zeta}{2\lambda - \cos\zeta}\right), \end{aligned} \tag{80}$$

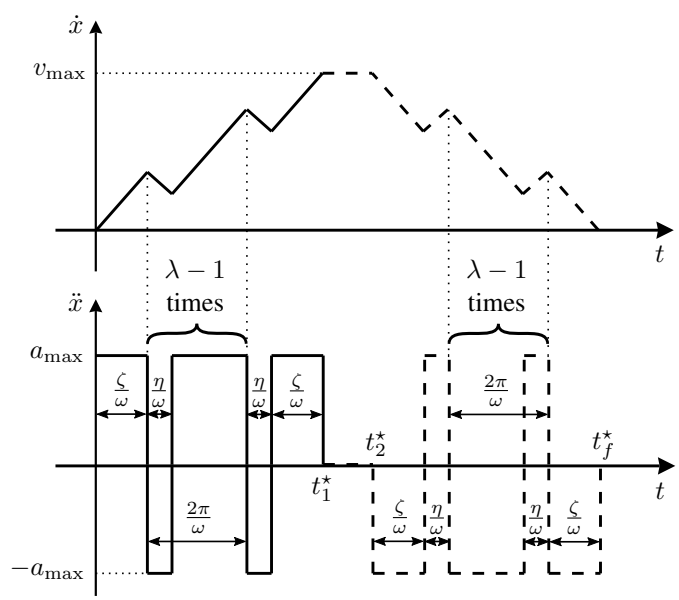

Fig. 11. Minimum-time vibration control profile of the transferring phase. Note that 2λ is the switching numbers (between $a_{\max}$ and $-a_{\max}$) of the control input segment spanning from $t = 0$ to $t = t_1^\star$, which is represented by the solid line.

where $K_m := (\omega v_{\max})/a_{\max}$, $0 \le \zeta \le \pi$, $0 \le \eta \le \pi/2$, and the index $\lambda \in \mathbb{N}^+$ is chosen such that following condition must be fulfilled

$$2\pi\lambda > K_m > 2(\lambda - 1)\pi. \tag{81}$$

In addition, the important switching instants of the bang-bang control input in Fig. 11 are given by

$$\begin{aligned} t_1^\star &= \frac{2\zeta + \eta + 2(\lambda - 1)\pi}{\omega} \\ t_2^\star &= \frac{\bar{x}_p - 2\int_0^{t_1^\star}\int_0^t \ddot{x}(\sigma)d\sigma dt}{v_{\max}} + t_1^\star. \end{aligned} \tag{82}$$

The minimum transferring time is $t_f^\star = t_1^\star + t_2^\star$.

The computation process of the minimum-time vibration control is explained as follows. Firstly, the index λ should be determined by using (81). Then, ζ and η can be calculated by solving the system of two nonlinear equations (80) through, for instance, *fsolve* routine of MATLAB. Next, the time instants $t_1^\star$, $t_2^\star$, and $t_f^\star$ can be computed by using (82). From the above information, the minimum-time vibration control profile can be established as suggested by Fig. 11.

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